STAT3888: Sensitivity Analysis of DSMKHHRB and MICE

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This document will do two things, one being a sensitivity analysis for the number of daily smokers variable `DSMKHHRB`, to see if it is worth analysing it more closely in causal inference terms, and two being an attempt to impute values using MICE and to see whether that can lead to robust analysis as well.

Also, by now I think we will mainly focus on the full population data set, since there is a chance that we may broaden the question anyway, and it makes things a lot cleaner.

`DSMKHHRB` Sensitivity Analysis

▼ Code

```
# Load dataset and models
fdat.full = readRDS(here("Objects", "fdat_full.rds"))
alt.fmlr = readRDS(here("Objects", "Full_data_MLR_ALT3.rds"))
ggt.fmlr = readRDS(here("Objects", "Full_data_MLR_GGT3.rds"))
```

Then, we simply just fit the models without `DSMKHHRB`, and see whether it makes a significant difference or not for MLR, and check convergence, AIC, and deviance.

▼ Code

```
# Fit models without DSMKHHRB
alt.fmlr.sub = multinom(formula = ALT3 ~ EXREGUIC + AGEB + SEX +
                        data = fdat.full, Hess = TRUE)
```

```
# weights:  60 (38 variable)
initial  value 5943.492482
iter  10 value 3781.752254
iter  20 value 3524.917763
iter  30 value 3489.908173
iter  40 value 3481.132097
iter  50 value 3475.442031
iter  50 value 3475.442008
iter  50 value 3475.442008
final  value 3475.442008
```

```
converged
```

▼ Code

```
ggt.fmlr.sub = multinom(formula = GGT3 ~ EXREGUIC + AGEB + SEX +
                        data = fdat.full, Hess = TRUE)
```

```
# weights:  60 (38 variable)
initial  value 5943.492482
iter  10 value 3358.043298
iter  20 value 3127.347342
iter  30 value 3077.827272
iter  40 value 3070.888935
iter  50 value 3069.613963
iter  60 value 3069.350151
final  value 3069.348577
converged
```

▼ Code

```
# Check convergence like in model diagnostics
conv = c(alt.fmlr.sub$convergence, ggt.fmlr.sub$convergence)
conv.log = (conv == 0)
models = c("ALT Without Smoking", "GGT Without Smoking")
aic = c(alt.fmlr.sub$AIC, ggt.fmlr.sub$AIC)
deviance = c(alt.fmlr.sub$deviance, ggt.fmlr.sub$deviance)
models.df = data.frame("Model" = models, "Convergence" = conv.lo
                       "AIC" = aic, "Deviance" = deviance)
kable(models.df, align = "cccc")
```

Model	Convergence	AIC	Deviance
ALT Without Smoking	TRUE	7026.884	6950.884
GGT Without Smoking	TRUE	6214.697	6138.697

It seems fine for now, so let us check if there are any changes to the significance of the coefficients, especially for `EXREGUIC`.

▼ Code

```
# Comparing ALT
stargazer(alt.fmlr, alt.fmlr.sub, type = "html",
          title = "DSMKHHRB Sensitivity Analysis for ALT",
          dep.var.caption = "ALT Level (Baseline: Low)")
```

	ALT Level (Baseline: Low)			
	Borderline	High	Borderline	High
	(1)	(2)	(3)	(4)
EXREGUICDid not meet recommended guidelines	-0.039	0.213**	-0.063	0.197**
	(0.092)	(0.090)	(0.091)	(0.089)
AGEB20-24 years	0.056	0.020	0.058	0.027
	(0.610)	(0.554)	(0.610)	(0.554)
AGEB25-29 years	0.699	0.159	0.719	0.167
	(0.551)	(0.514)	(0.550)	(0.514)
AGEB30-34 years	0.459	-0.034	0.476	-0.025
	(0.553)	(0.513)	(0.553)	(0.513)
AGEB35-39 years	0.126	0.318	0.151	0.330
	(0.558)	(0.502)	(0.557)	(0.502)
AGEB40-44 years	0.287	0.048	0.305	0.056
	(0.548)	(0.502)	(0.547)	(0.502)
AGEB45-49 years	0.702	0.281	0.728	0.288
	(0.541)	(0.498)	(0.540)	(0.498)
AGEB50-54 years	0.766	0.272	0.786	0.282
	(0.540)	(0.498)	(0.540)	(0.498)
AGEB55-59 years	0.394	-0.004	0.418	0.004
	(0.544)	(0.500)	(0.543)	(0.500)
AGEB60-64 years	0.654	-0.099	0.695	-0.076
	(0.539)	(0.499)	(0.539)	(0.499)
AGEB65-69 years	0.122	-0.352	0.172	-0.324
	(0.548)	(0.505)	(0.548)	(0.505)
AGEB70-74 years	0.088	-0.924*	0.155	-0.886*
	(0.557)	(0.528)	(0.556)	(0.527)
AGEB75-79 years	-1.152*	-1.528***	-1.097*	-1.498***
	(0.628)	(0.576)	(0.628)	(0.575)
AGEB80-84 years	-0.725	-3.555***	-0.652	-3.510***
	(0.631)	(1.115)	(0.630)	(1.115)
AGEB85 years and older	-2.388**	-3.039***	-2.321**	-3.003***
	(1.139)	(1.120)	(1.139)	(1.119)
SEXMale	1.010***	0.809***	1.010***	0.809***
	(0.093)	(0.090)	(0.093)	(0.090)
WAISTNot at risk	-0.514***	-0.838***	-0.521***	-0.840***
	(0.134)	(0.149)	(0.134)	(0.149)

WAISTSubstantially increased risk	0.461***	0.806***	0.450***	0.800***
	(0.113)	(0.112)	(0.113)	(0.112)
DSMKHHRBOne	-0.210	-0.161		
	(0.134)	(0.125)		
DSMKHHRBTwo	-0.545*	-0.161		
	(0.280)	(0.225)		
DSMKHHRBThree or more	0.506	-0.014		
	(0.685)	(0.799)		
Constant	-2.839***	-2.476***	-2.896***	-2.510***
	(0.540)	(0.498)	(0.540)	(0.498)
Akaike Inf. Crit.	7,030.810	7,030.810	7,026.884	7,026.884

Note: *p<0.1;* ***p<0.05;*** p<0.01

DSMKHHRB Sensitivity Analysis for ALT

▼ Code

```
# Comparing GGT
stargazer(ggt.fmlr, ggt.fmlr.sub, type = "html",
          title = "DSMKHHRB Sensitivity Analysis for ALT",
          dep.var.caption = "GGT Level (Baseline: Low)")
```

	GGT Level (Baseline: Low)			
	Borderline	High	Borderline	High
	(1)	(2)	(3)	(4)
EXREGUICDid not meet recommended guidelines	0.164	0.361***	0.212*	0.396***
	(0.123)	(0.086)	(0.122)	(0.085)
AGEB20-24 years	12.230***	0.605	11.537***	0.583
	(0.373)	(1.110)	(0.372)	(1.109)
AGEB25-29 years	12.281***	1.046	11.576***	1.020
	(0.253)	(1.043)	(0.252)	(1.042)
AGEB30-34 years	12.178***	1.529	11.481***	1.489
	(0.251)	(1.031)	(0.251)	(1.030)
AGEB35-39 years	12.230***	1.479	11.525***	1.426
	(0.223)	(1.028)	(0.222)	(1.027)
AGEB40-44 years	12.251***	1.769*	11.555***	1.726*
	(0.206)	(1.023)	(0.206)	(1.022)
AGEB45-49 years	12.436***	1.896*	11.733***	1.852*
	(0.184)	(1.021)	(0.184)	(1.020)

AGEB50-54 years	12.517***	2.239**	11.809***	2.192**
	(0.182)	(1.020)	(0.182)	(1.019)
AGEB55-59 years	12.437***	2.042**	11.735***	2.002**
	(0.182)	(1.020)	(0.181)	(1.019)
AGEB60-64 years	12.552***	2.264**	11.808***	2.184**
	(0.168)	(1.019)	(0.168)	(1.018)
AGEB65-69 years	12.280***	2.222**	11.520***	2.135**
	(0.203)	(1.021)	(0.203)	(1.019)
AGEB70-74 years	12.339***	1.939*	11.546***	1.827*
	(0.231)	(1.026)	(0.230)	(1.024)
AGEB75-79 years	12.170***	2.037**	11.403***	1.935*
	(0.288)	(1.030)	(0.288)	(1.028)
AGEB80-84 years	12.524***	1.803*	11.704***	1.680
	(0.304)	(1.041)	(0.303)	(1.040)
AGEB85 years and older	11.878***	2.101**	11.090***	1.989*
	(0.559)	(1.057)	(0.558)	(1.056)
SEXMale	0.321***	-0.023	0.318***	-0.023
	(0.121)	(0.084)	(0.121)	(0.084)
WAISTNot at risk	-0.710***	-0.504***	-0.697***	-0.501***
	(0.191)	(0.138)	(0.191)	(0.137)
WAISTSubstantially increased risk	0.299**	0.555***	0.316**	0.556***
	(0.147)	(0.106)	(0.147)	(0.106)
DSMKHHRBOne	0.508***	0.212*		
	(0.154)	(0.116)		
DSMKHHRBTwo	0.484*	0.622***		
	(0.294)	(0.199)		
DSMKHHRBThree or more	-11.248***	0.864		
	(0.00000)	(0.599)		
Constant	-15.349***	-4.162***	-14.554***	-4.056***
	(0.152)	(1.020)	(0.151)	(1.018)
Akaike Inf. Crit.	6,202.484	6,202.484	6,214.697	6,214.697

Note: *p<0.1;* ***p<0.05;*** p<0.01

DSMKHHRB Sensitivity Analysis for ALT

It does seem to slightly change the significance level for GGT Borderline `EXREGUIC`, although only up to $\alpha = 0.1$, so I think it is decently robust without `DSMKHHRB`. In this case, I think we may want to drop it from our models **if we decide that either smoking does not have an association with liver biomarkers (Question 1) or if this particular variable is not a confounder for**

physical activity either.

MICE Imputation

I want to try and do the MICE imputation from lecture, and use that to impute missing data values in our dataset, and see if that can balance out our classes more and essentially mitigate the issue of sparsity we had earlier on; we also will do sensitivity analysis on the ALT and GGT models we had earlier as well.

▼ Code

```
# Impute using MICE; note we have to remove the ID variables fo
set.seed(102)
fdat.noid = fdat.full %>% select(-c(ABSLID, ABSPID))

file.path = here("Objects", "fdat_imputed.rds")
if (file.exists(file.path)) {
  fdat.imp = readRDS(file.path)
} else {
  fdat.imp = mice(fdat.noid, m = 20, maxit = 10, method = "pmm",
  saveRDS(fdat.imp, file.path)
}
```

Then, we fit the model with the full data, using the specialised `mice` functions. We just take the main variable `EXREGUIC` and its significance for now, since the output is quite a handful so I don't think `stargazer` will work as cleanly here.

▼ Code

```
# MICE for ALT
alt.fmlr.mice = with(fdat.imp,
                     expr = multinom(formula = ALT3 ~ EXREGUIC +
                                       SEX + WAIST + DSMKHHRB,
alt.fmlr.mice.pooled = pool(alt.fmlr.mice)
df = data.frame(summary(alt.fmlr.mice.pooled))
alt.exreguic.rows1 = df[c(2, 24), ] %>%
  mutate("y.level" = c("Borderline", "High")) %>%
  relocate(last_col()) %>% select(-c(term, df))
alt.exreguic.rows2 = tidy(alt.fmlr)[c(2, 24), ] %>%
  select(-c(term))
alt.rows = rbind(alt.exreguic.rows1, alt.exreguic.rows2) %>%
  mutate("Model" = rep(c("MICE", "Original"), each = 2)) %>%
  relocate(last_col())
rownames(alt.rows) = NULL
kable(alt.rows, caption = "Sensitivity Analysis for MICE on ALT'
```

```
      digits = 4, align = "cccccc",
      col.names = c("Model", "ALT Level", "Coefficient",
                    "SE", "t-Score", "p-value"))
```

Sensitivity Analysis for MICE on ALT

Model	ALT Level	Coefficient	SE	t-Score	p-value
MICE	Borderline	-0.0204	0.0695	-0.2935	0.7697
MICE	High	0.0746	0.0630	1.1837	0.2383
Original	Borderline	-0.0391	0.0922	-0.4245	0.6712
Original	High	0.2128	0.0898	2.3684	0.0179

We do the same for GGT.

▼ Code

```
# MICE for GGT
ggt.fmlr.mice = with(fdat.imp,
                     expr = multinom(formula = GGT3 ~ EXREGUIC +
                                       SEX + WAIST + DSMKHHRB, t
ggt.fmlr.mice.pooled = pool(ggt.fmlr.mice)
df = data.frame(summary(ggt.fmlr.mice.pooled))
ggt.exreguic.rows1 = df[c(2, 24), ] %>%
  mutate("y.level" = c("Borderline", "High")) %>%
  relocate(last_col()) %>% select(-c(term, df))
ggt.exreguic.rows2 = tidy(ggt.fmlr)[c(2, 24), ] %>%
  select(-c(term))
ggt.rows = rbind(ggt.exreguic.rows1, ggt.exreguic.rows2) %>%
  mutate("Model" = rep(c("MICE", "Original"), each = 2)) %>%
  relocate(last_col())
rownames(ggt.rows) = NULL
kable(ggt.rows, caption = "Sensitivity Analysis for MICE on GGT"
      digits = 4, align = "cccccc",
      col.names = c("Model", "GGT Level", "Coefficient",
                    "SE", "t-Score", "p-value"))
```

Sensitivity Analysis for MICE on GGT

Model	GGT Level	Coefficient	SE	t-Score	p-value
MICE	Borderline	0.1312	0.0899	1.4591	0.1477
MICE	High	0.1886	0.0632	2.9868	0.0033
Original	Borderline	0.1642	0.1233	1.3317	0.1829

Original	High	0.3610	0.0856	4.2187	0.0000